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Technical paper

Sensor placement utilizing a digital twin for thermal error compensation of machine tools

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ABSTRACT

Thermal errors in machine tools significantly impact precision and, therefore, productivity. Mitigating these errors often results in a trade-off between energy efficiency and accuracy. While data-driven compensation models show promise in addressing this challenge and achieving sustainable precision, their effectiveness hinges on the careful selection and placement of sensors as model inputs. This paper introduces a novel temperature sensor positioning method for thermal error compensation that leverages a digital twin framework to virtually determine ideal sensor positions and their effects on the compensation model. By accurately identifying temperature-sensitive points, our approach improves compensation accuracy and reduces the number of sensors required, thus enhancing both model robustness and operational efficiency. For choosing this set not only one simulation model is used but an ensemble with varying boundary conditions and thus model properties. Validation results show that the proposed method outperforms traditional, manually determined sensor placement strategies, providing a scalable solution for adaptable, energy-efficient thermal management in precision manufacturing. The selected sensor set based on a hybrid singular value decomposition and Least Absolute Shrinkage and Selection Operator approach yields a more robust compensation using only 7 instead of the manually chosen 22 temperature sensors. The thermal error reduction ranges from 77%–94% using simulated data with a corresponding reduction of 75%–85% achieved on the physical machine.

1. Introduction

Thermal errors in machine tools (MTs) remain one of the largest obstacles to sustainable and high-precision production. According to Mayr et al. [1] up to 75% of workpiece errors are induced by the effects of temperature changes. While much research has been done on mitigating these effects, the challenge of robust and efficient compensation eludes widespread industrial use, though emerging digital twin technologies show promise in overcoming these limitations [2]. Up to 50% of scrap production can be caused by thermal errors according to a survey done by Putz et al. [3]. In order to achieve the precision requirements demanded by industry, MTs are typically subject to extensive thermal stabilization. Denkena et al. [4] concluded that the cooling systems of MTs typically require between 15% and 30% of their total energy consumption. Advances have been made to improve the energy efficiency of MTs while accounting for their thermal behavior. For example, Denkena et al. [5] reviewed various spindle cooling concepts which require a large effort if the thermomechanical effects are to be minimized. Mori et al. [6] investigated the trade-off between energy efficiency of MTs and accuracy not only for individual components but the entire factory hall. Besides additional design and tempering measures, the thermal error can also be compensated in real-time using the MT controls, which is becoming a popular approach due to its potential for achieving sustainable accuracy [7]. For compensation, both physical and data-driven models can be utilized. Physical models such as those based on the Finite Element Method (FEM) typically require a suitable system representation and are often computationally expensive [8]. For real-time applications, often data-driven models are utilized [9,10], for example, using a Python implementation as demonstrated by Mares et al. [11]. Ma et al. [12] used a neural network approach based on long short-term memory (LSTM) to transfer models trained for thermal compensation at different spindle speeds. On-machine measurements are state of the art for kinematic calibrations and new methods of combining these with machine learning are being developed according to Gao et al. [13]. This paper introduces a novel sensor positioning method that leverages a digital twin to virtually define the optimal sensor placements for robust and efficient thermal error compensation.

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1.1. Selection of model inputs for thermal compensation of machine tools

Temperature-sensitive point or model input selection is a crucial step in the application of thermal error compensation models. It is differentiated from sensor positioning as it only chooses a subset from a larger set of sensors, typically based on physical sensors installed on a MT. Researchers have explored various approaches of sensor selection to choose suitable model inputs out of the available measurements, aiming to increase both robustness and accuracy in compensation models. Czwartosz and Jedrzejewski [14] analyzed 40 publications from 2012 to 2022 and found studies containing some form of input selection to, on average, achieve higher compensation accuracy compared to those without. Liu et al. [15] identified the selection of temperature sensors to be the main factor influencing the robustness of the compensation model as the collinearity between the sensors can lead to undesirable results if not managed properly. Abdulshahed et al. [16] used an infrared camera and fuzzy c-means clustering to identify optimal sensor locations. Li et al. [17] used grey system theory to reduce the temperature inputs in a model from 16 to 4, increasing robustness and prediction accuracy.

Blaser et al. [18] introduced a closed-loop thermal error compensation approach with retraining of the model parameters if the accuracy is no longer sufficient. Zimmermann et al. [19] extended this with adaptive input selection which enables the model to adapt its structure to new situations not available in the initial training data set. Group LASSO (Least Absolute Shrinkage and Selection Operator) can enhance this approach as it optimizes both the model parameters and structure in the same step [20]. Ye et al. [21] used the Group LASSO for input selection and predicted the thermal error with a different approach based on extreme gradient boosting. Wei et al. [22] extended this approach with ARX and GPR models and showed that it outperformed state-of-the-art deep learning approaches. This shows the efficiency of data-driven modeling approaches and the relevance of input selection for their efficiency, but the question of which sensor to consider for input selection in the first place and their optimal placement remains open.

1.2. Positioning of temperature sensors

The positioning of sensors happens before any are physically installed and so is not a reductive but rather a generative process. A set is not shrunk but generated which is later placed on the actual machine. For doing this autonomously simulations have to be used that allow a consideration of the whole MT. ISO 230-3 [23] makes general recommendations for positions of temperature sensors based on “engineering knowledge”, such as close to heat sources, the position transducers or the ambient temperature. Holub et al. [24] confirmed the ISO 230-3 [23] recommendation that placing sensors near the axes of motion and linear encoders is essential. They also demonstrated that poorly positioned temperature sensors significantly reduce the accuracy of thermal error compensation. Du et al. [25] considered a 1D structure of spindles and investigated the optimal sensor location for one temperature sensor, which was successfully validated using a real spindle of an MT. Straka et al. [26] investigated the impact of the positioning of temperature sensors for the model transfer of different milling MTs. Zhan et al. [27] used fuzzy c-medoid clustering to optimize temperature sensor placement for subsequent thermal error prediction of a spindle bearing system. Sa et al. [28] used a genetic algorithm to iteratively improve the temperature sensor positions of a spindle component based on a FEM model. Kizaki et al. [29] used a FEM simulation to place 284 sensors in locations with large temperature gradients, which were used to directly measure the entire temperature field using linear interpolation. Teshima et al. [30] extended this approach using Model Reduction inside ANSYS to reduce the computational demand and using the sensitivity of each node to the thermal displacement to choose the most sensitive nodes. The grey relational grade can also be used to identify the most dissimilar sensors in FEM for compensation with a LSTM model as shown by Liang et al. [31]. Ultimately, these studies collectively demonstrate that efficient sensor positioning is vital for accurate thermal error compensation in MTs. Intelligent sensor placement enhances model reliability as well as accuracy and reduces cost and complexity, instigating the need for a holistic approach considering a digital twin. However a combination with the compensation model is missing and considering the entire machine tool, especially under varying boundary conditions, remains an unsolved challenge.

1.3. Thermal modeling of machine tools

In the manufacturing value chain, digital twins offer promising integration of simulations with process control and monitoring [32]. However, leveraging full-scale finite element (FE) models is often impractical due to computational limitations. To address this, model order reduction techniques have become essential for enhancing computational efficiency [33,34]. Irino et al. [35] have successfully used a model order reduced digital twin for the real-time compensation of both thermal and process force-induced errors of an MT. An exemplary application by Gomez-Acedo et al. [36] employs a Kalman filter and a lumped mass model with four states, creating a simple physics-based compensation model. Lang et al. [37] use a Kalman filter with k-means clustering and correlation-based input selection to estimate the thermal state of an entire MT in order to use it for subsequent thermomechanical simulation and thermal error compensation. The use of singular value decomposition together with an unscented Kalman filter to identify the translational thermal error model of a MT spindle solely based on measurement data was shown by Fu et al. [38]. Fujishima et al. [39] employed a deep learning approach to create an end-to-end model mapping temperature data to thermal errors, identifying sensor failure as a critical risk because erroneous measurements can destabilize compensation. Consequently, their findings underscore the importance of minimizing the number of sensors to reduce both potential failure points and overall system complexity.

This paper aims to close the gap between current research in thermal simulations and digital twins as well as thermal error compensation. By leveraging the possibility of a simulation ensemble, different approaches for finding an ideal sensor set for thermal error compensation are proposed and developed. This reduced the cost and complexity of thermal compensation significantly as less but more effective sensors can be used which also reduced the risk of failure. First, an overview of the thermomechanical simulations of the investigated grinding MT is given, which serves as the foundation for the digital twin. Subsequently, the different approaches for selecting a suitable sensor set for thermal error compensation are introduced before the compensation model and on-machine measurements are introduced. Finally, the results compare the different sensor sets and show a validation of the best-performing sensor set on the physical MT to show the applicability and benefit in a real production setting before concluding and giving an outlook to future work.

2. Thermomechanical simulation model as basis for a digital twin

Initially, the process involves utilizing a Computer-Aided Design (CAD) model to create a FE mesh for each component using the commercial software ANSYS, which is commonly used for such tasks [40]. Following this, the mesh is exported to the MORe software [41,42], to treat the mesh as a state space representation for both thermal and mechanical systems. Subsequently, the model is compressed into a smaller dimensional space through Krylov and Modal Subspace (KMS) reduction. This method maintains precision within the specified frequency range, as demonstrated by Hernández-Becerro et al. [43] for thermal systems and Spescha et al. [44] for mechanical and mechatronic systems. Fig. 1 summarizes the workflow of the simulation setup described above as the basis for the digital twin representation of the

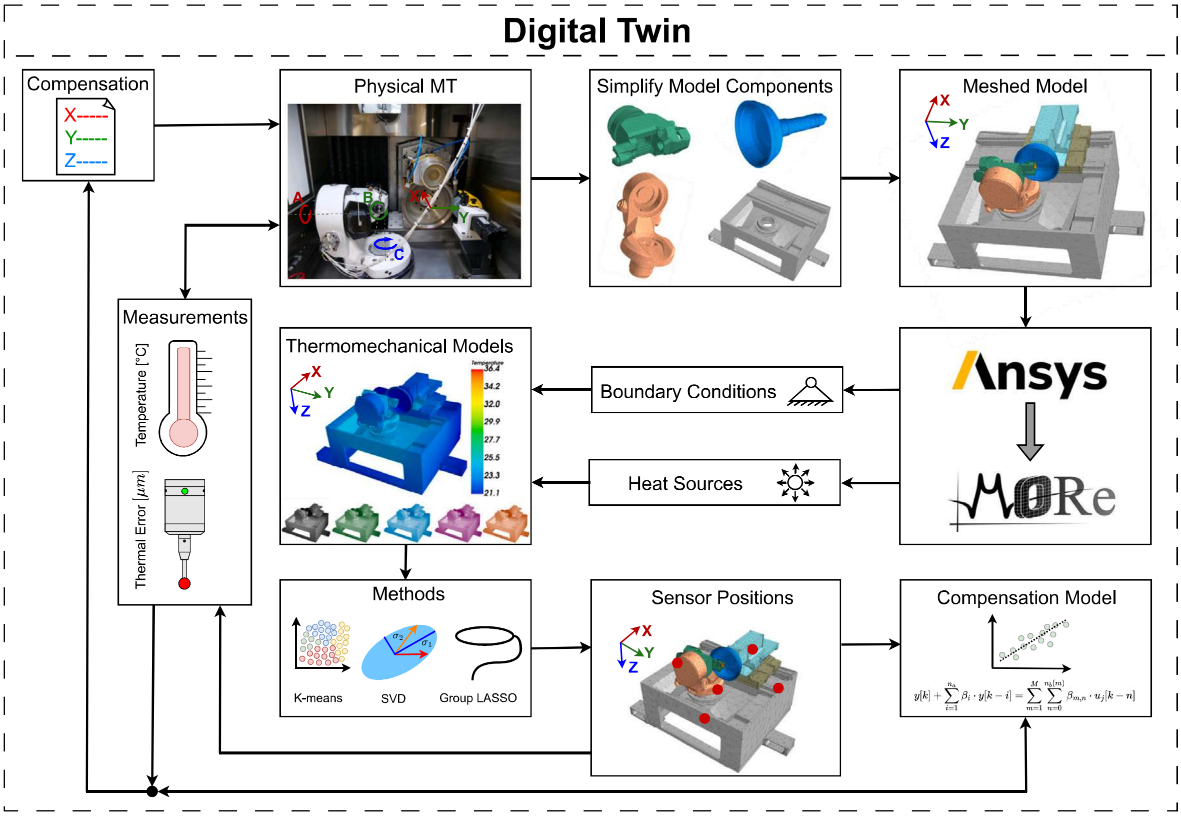

Fig. 1. Overview of the development of the thermomechanical model and its integration in the digital twin of the MT. Through the variation of the model properties and boundary conditions different thermomechanical models can be generated to represent different machine behavior types that can occur in differing conditions upon which the sensor positioning is done.

physical MT. The thermomechanical model is used to select suitable sensor sets using different methods that are introduced in Section 3. These sensor positions are then measured either on the simulation model or for the physical MT to be used with the compensation model. The compensation model is trained using measurements from the actual machine or from the validation simulation. The predicted values can then be fed back to the machine control to adjust it for the specific thermal state.

The properties of the models used for thermal analysis are summarized in Table 1. The reduced thermomechanical model has 923 degrees of freedom. The dynamics of the thermal system are represented in Eq. (1) as a linear time-invariant (LTI) system, which can be used in a thermal system to represent a first-order system of ordinary differential equations (ODE).

$$E\dot{x}(t) = Ax(t) + Bu(t) \tag{1}$$

A is the system matrix that includes conduction and convection effects, while E is the mass matrix that represents the thermal capacity. The input matrix B and the input vector $u(t)$ are part of the model. The thermal state $x(t)$ usually represents the temperature distribution in the system. The physical parameters such as material properties and convective heat transfer coefficients are assumed to be constant. For the mechanical aspect of the digital twin, the dynamics are described in Eq. (2).

$$y_{mech} = C_i K^{-1} K_{th} x(t) \tag{2}$$

In this Equation, $y_{mech}(t)$ represents the mechanical deformation at time t. The term C_i enables the mechanical coupling of different parts of the system, such as the tool center point (TCP) and the workpiece center. The mechanical stiffness matrix K^{-1} is derived from the material properties and the system mesh, while K_{th} denotes the coupling matrix that connects the thermal and mechanical systems. This coupling matrix introduces a volumetric force due to thermal expansion, which is influenced by the thermal expansion coefficient α.

2.1. Modeling of the grinding machine tool

The investigation of optimal sensor positions for thermal error compensation is applied on a 5-axis precision grinding MT from Agathon AG. This MT is equipped with an actively cooled metal working fluid (MWF) supply, as well as separate cooling circuits for the drives and structure. The MWF, which is typically a cutting oil, in this case is Blasogrind GTM 4. Which helps to dissipate the generated heat during the grinding operation. This also distributes some of that heat to the structure of the MT and has generally a strong influence on the thermal characteristics of grinding MTs [45]. MWF is ejected from eight nozzles, in total four on both sides of the grinding wheel facing down to the process zone. The X- and Y-axes of the MT are designed as cross-slides and are separated from the influence of the grinding chamber with lamella bellows, which splits up the enclosure of the MT into two sections. These axes are driven by linear motors and controlled by linear encoders, also all rotary axes have encoders. These encoders are considered in the simulation model with a closed-loop position control. The spindle is mounted on the X-axis slide which carries the grinding wheel. The B-axis represents the rotational axis of the clamped workpiece. Additionally, the whole B-axis can be rotated around the A-axis which is positioned on the C-axis structure. The C-axis denotes the vertical axis of rotation for the work headstock and is located on the base component.

The thermomechanical model is reduced using the tool MORe [41]. MORe enables model reduction approaches to reduce the computational cost of thermomechanical simulations significantly and is especially suited for the simulation of MTs. One key advantage is that it can consider the position-dependent MT behavior with damping without

Table 1
Number of DOFs of the components of the thermal model before and after the implementation of the model reduction technique.

Component	#DOFs system	#DOFs reduced
X-Axis	468,798	89
Y-Axis	121,764	246
A-Axis	111,900	82
B-Axis	45,687	75
C-Axis	132,786	78
Spindle	121,269	75
Basis	168,549	278
Total	1,170,753	923

Table 2
HTC values for specific components applied in the simulation. High HTC values are attributed to the forced convection from the MWF or cooling system interacting with the surfaces, enhancing the heat transfer.

Component	HTC $[W/m^2K]$
X-Axis	5–6
Y-Axis	6–8
C-Axis	11–18
Basis	12–16
Spindle	210–450
Cooling	1283–1334
MWF	1271–1556

requiring a re-reduction or re-meshing of the model, as shown by Semm et al. [46]. The thermomechanical model consists of a X-, Y-, A-, B-, C-axis and a base and spindle component. Certain complex structural elements of the model are simplified for the digital twin, such as the shape of the bearings, holes, and edges, as these do not influence the thermomechanical behavior and can be abstracted by their dynamic characteristics. This ensured that the subsequent mesh is well conditioned and singularities are removed. Additionally, some components like motors are eliminated and replaced with heat sources and equivalent dynamic mass. The cooling system of the MT is considered by adding convective interfaces to simulate cooling effects around drives and cooling channels. The materials are defined according to the provided information from the manufacturer. The Ansys meshing tool is used on the setting coarse with adaptive sizing for generating the thermomechanical model, as MORe allows for further model order reduction at a later stage to reduce the computational time of the simulation. The mesh is generally unstructured, which makes it ideal for handling complex geometries. Tetrahedral elements with quadratic element order are employed to generate the mesh for the components. Furthermore, the mesh is refined in critical areas, including the B-axis clamping pin, in order to guarantee the accurate representation of the geometry of complex key components of the MT and prevent a skewness above 0.5. The DOFs of the model components are reduced through the application of the model order reduction (MOR) technique, as provided by MORe using Krylov and modal subspace reduction. A summary of the reduced number of DOFs is provided in Table 1.

In grinding MTs, the use of cutting fluid is an indispensable part of the production process which is crucial for absorbing heat created during the machining [47]. This cutting fluid, however, also has a significant effect on the thermal errors of the MT. The employment of cutting fluid leads to significant changes in the temperature distribution depending not only on the setup and properties of the MT but also the machining position, as shown by Shi et al. [48]. Distributed interfaces on locations where the cutting fluid is expected to flow are defined to capture the forced convective behavior of the cutting fluid in contact with the MT surface.

Only the structure of the MT, without the enclosure or full Computational Fluid Dynamics (CFD) is modeled. The aim of the DT is not to obtain an exact representation of the MT, but to reproduce the characteristic behavior of the MT in relation to temperature distribution and displacement.

The DT is used to perform thermomechanical simulations to obtain simulated temperature as well as the displacement data between the TCP and workpiece. The temperature is measured from all surface nodes contained in the FEM model. Therefore, around 185,000 measurement points of the surface temperature field of the MT are obtained. A data basis like this can only be generated using DTs since a very granular representation of the thermal state is obtained. A similar snapshot cannot be experimentally measured because of the prohibitive cost and effort required. Using such a large number of physical temperature sensors or thermal imaging systems on all surfaces introduces additional challenges, as the thermal behavior is complex.

2.2. Simulation of the thermomechanical machine tool behavior

To effectively capture the thermomechanical behavior of the 5-axis grinding MT, it is crucial to have representative simulation data. The objective of varying the boundary conditions is to introduce variations in the datasets since these MTs are used in a wide variety of production scenarios, and the MT is expected to maintain a consistent accuracy regardless. In this context, a 308-h load case has been simulated, consisting of five distinct segments, as shown in Fig. 2. Each segment contains a unique combination of heat sources with different heat loads, properties of convective interfaces, and temporal loads. Essentially, the load case includes different manufacturing processes of the MT, with both shorter and longer exposure of heat loads. The fourth load case depicted in Fig. 2 illustrates the behavior of an active MT without an active grinding process, essentially idling in the ready-to-operate state, such as during a setup change. In this case, the heat sources continuously produce a constant amount of heat. This setup allows to observe how changes in the ambient temperature affect the active but idle system. The magnitudes of the different load cases of heat sources vary in the range of $\pm 15\%$. The variations of the convective heat transfer coefficients (HTC) applied in the simulations are summarized in Table 2. This variation of the model parameters reflects the uncertainty associated with defining them and allows the simulations to cover a broad range of underlying behavior. In reality these parameters often change slightly during operation. Varying these parameters therefore allows the selection of sensor positions that are robust to slight changes in boundary conditions and are optimal for a broad range of MT uses. Other model parameters are not varied such as the thermal contact conductivity (TCC), which was chosen to be 100 W/K for the linear rails and 3–18 W/K for the rotary axis bearings depending on their size and characteristics. This choice was carried through according to best practices following Hernández-Becerro [49]. This results in the segments are characterized by rapid oscillations of varying amplitude. Conversely, other sections show gradual changes in displacement values over time due to different boundary conditions within the load cases. The temperature measurements are collected from all surface nodes of the FEM model and the displacement between the TCP and the workpiece in the X-, Y- and Z-directions are recorded from carrying through the mechanical simulation based on the thermal state.

As this data, based on five different thermomechanical models, is used to select the optimal temperature sensor positions, it should not be used to benchmark the resulting compensation models. Therefore, an additional simulation is carried through with different, still varying, parameters and properties that differ from the previous simulations, so creating a new thermomechanical model, to serve as a validation data set. This validation simulation is conducted for a continuous period of 335 h, as illustrated in Fig. 3. Ensuring a smooth and uninterrupted simulation is required to validate the predictive capabilities of the compensation model for thermal errors which are trained on the first half of this validation simulation and evaluated on the second part. To introduce realistic environmental impacts into the simulation, a sine curve with a period of 24 h is used for the environmental temperature. The mean temperature is set at 20.5 °C with an amplitude of 1 °C.

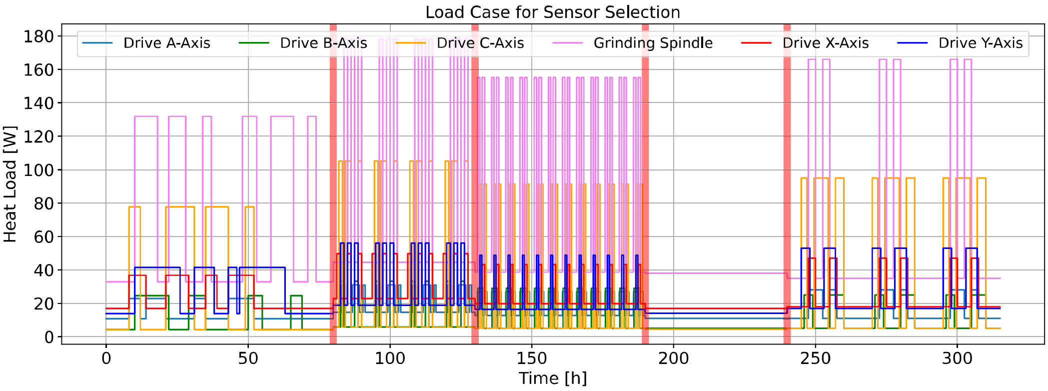

Fig. 2. Load case of 308-h duration for the evaluation of sensor positions. It consists of five distinct machining conditions with changing thermal boundary conditions. The red areas represent an interruption and restart of the simulation to start at a new initial state.

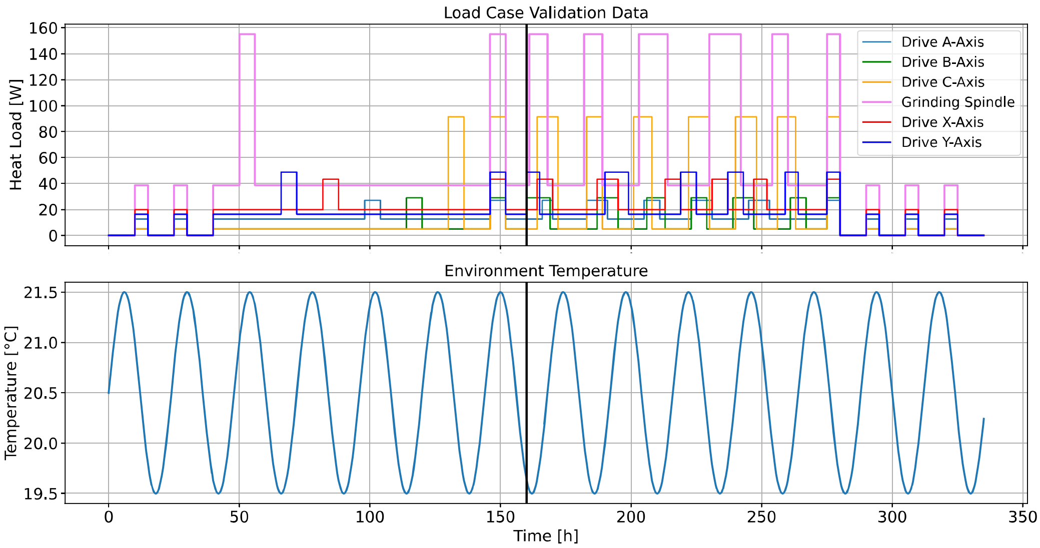

Fig. 3. Continuous load case of 335 h applied to the simulation model of the DT. This is used as the second data set so for the generation of an ARX compensation model to validate the temperature sensors selected based on the first data set. The black line after 160 h represents the split into a training and test set for the model.

This integration enables the simulation to mimic the day-night cycle, effectively capturing idealized temperature fluctuations commonly encountered in real-world scenarios. The aim of using this dataset is to assess the performance in both training and validation of compensation models based on the chosen sensor positions using a different dataset than the one used for the selection of sensor positions.

All Simulations were carried out on a desktop workstation using an Intel i9-13900KS 24 core processor. To generate the validation data set which corresponds to 335 h, the simulation takes 2 h 37 min and 49 s including the time for reducing the model once. This means the simulation is more than 128 times faster than real measurements. The power consumption of the entire workstation was measured using a Voltcraft SEM6000 device, totaling 0.452 kWh, which corresponds to an average power draw of 173 W during the simulation, If this is compared to the MTs powerdraw of 3–5 kW plus 2–3.5 kW for the MWF tempering and processing the workstation takes roughly ∼37 times less power. When considering energy consumption per measurement-equivalent hour, the simulations require ∼4815 times less energy than real measurements, making them a significantly more sustainable alternative for generating data. This estimate does not even account for the additional requirements of the physical machine, such as compressed air and the much higher capital costs compared to a simulation workstation. Another advantage of the simulation is the fact that arbitrary boundary conditions can be applied as well as the temperature of any mesh element recorded which allows for the selection of the ideal temperature sensor position. Using the much more costly experimental data only input selection can be carried through, so choosing the best of the existing sensors. However, it is important to note that experiments still provide a higher level of trust in fidelity and remain essential for validating the simulation approach.

3. Temperature sensor positioning approaches for thermal compensation

This section discusses different approaches for sensor positioning. To this end, the required methods, such as k-means clustering, singular value decomposition (SVD), and LASSO, are introduced and subsequently applied to the first simulation dataset.

3.1. Engineering knowledge based manual sensor positioning

In order to define a baseline for the sensor positioning as well as validate the DT with experimental data, a reference set of sensor positions is required. For this, a manual positioning of the temperature sensors can be based on engineering knowledge of the designers and best practices considering the recommendations of ISO 230-3 [23]. These recommendations encompass positioning sensors in close proximity to pertinent thermal heat sources within the MT, as well as strategic placement on various components such as the cooling system, and monitoring ambient temperature fluctuations. The positioning process

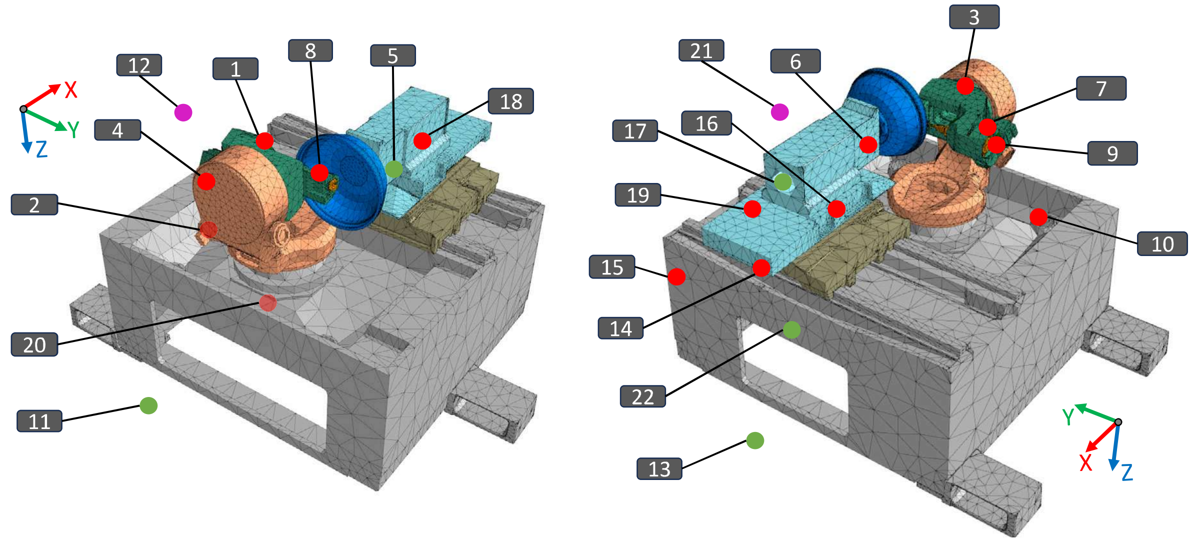

Fig. 4. Temperature sensors manually positioned based on engineering knowledge. Red dots indicate sensors represented on the structural components of the digital twin (DT). Green dots mark sensors that cannot be represented in the DT due to missing components, such as process fluids or system boundaries. Violet dots represent enclosure temperature sensors modeled as boundary conditions influenced by the surrounding environment boundary condition. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

involved direct evaluation of these proposed locations on the physical MT, followed by a translation into the DT. However, inherent trade-offs emerged during this translation phase, stemming from constraints such as the absence of specific structural components, thereby precluding the accommodation of all sensors on the DT. For example, the temperatures of the MWF, the coolant, and the ambient environment are measured experimentally. These measurements are set as boundary conditions in the DT simulations, as no CFD components are explicitly modeled, and modeling beyond the MT core components is not considered as it is currently not feasible. Fig. 4 visualizes the positions of sensors proposed and subsequently installed and modeled based on the engineering knowledge. A total of 22 temperature sensors are installed on the physical MT. Due to limitations in the simulation, such as no environmental or MWF temperature as these are not modeled but set as boundary conditions, a total of 17 sensors are included in the simulation. These are all sensors on structural parts of the machine.

3.2. Using clustering centroids as sensor positions

A first straightforward approach for sensor positions is to separate the entire MT into different regions and choose the representative center of each of these regions as a temperature sensor position. For this, clustering analysis is a suitable technique for generating effective groups from simulation data with different attributes. An often used [51] clustering method is the K-means clustering algorithm, which aims to find K number of clusters in a given data set [52]. The K-means algorithm starts by randomly selecting centroids as initial cluster centers. Then, each data point is assigned to the nearest centroid based on Euclidean distance. Subsequently, new cluster centers are calculated by taking the average of the points in each cluster. This process repeats until no further changes in cluster assignments occur. The K-means algorithm is mathematically described as follows:

$$\underset{\mathbf{S}}{\arg\min} \sum_{i=1}^{k} \sum_{\mathbf{x}\in S_i} \|\mathbf{x}-\boldsymbol{\mu}_i\|^2 = \underset{\mathbf{S}}{\arg\min} \sum_{i=1}^{k} |S_i| \operatorname{Var} S_i \tag{3}$$

Each cluster S_i represents a group of temperature sensors that have similar temperature patterns over time. The total number of distinct clusters is denoted by k. The centroid of each cluster S_i, denoted by μ_i, is defined as the average temperature time series for all sensors belonging to that cluster. A sensor has a corresponding time series of temperature readings, represented by a vector $\mathbf{x}$. Sensors are assigned to the cluster with the centroid μ_i that is closest to their temperature time series. The assignment is based on the Euclidean distance, which measures the similarity between the sensor's temperature time series and the centroid. Once all sensors have been assigned to their respective clusters, the centroids are recalculated as the mean of the temperature time series for all sensors within each cluster. This process of assigning sensors to clusters and updating centroids is repeated iteratively. The procedure continues until the clusters stabilize; in other words, the

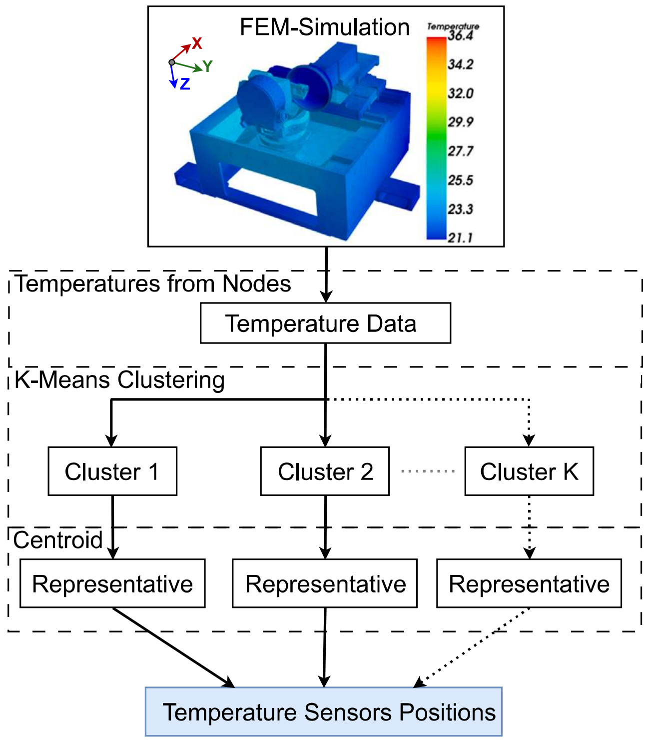

Fig. 5. Flow chart of centroid approach. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

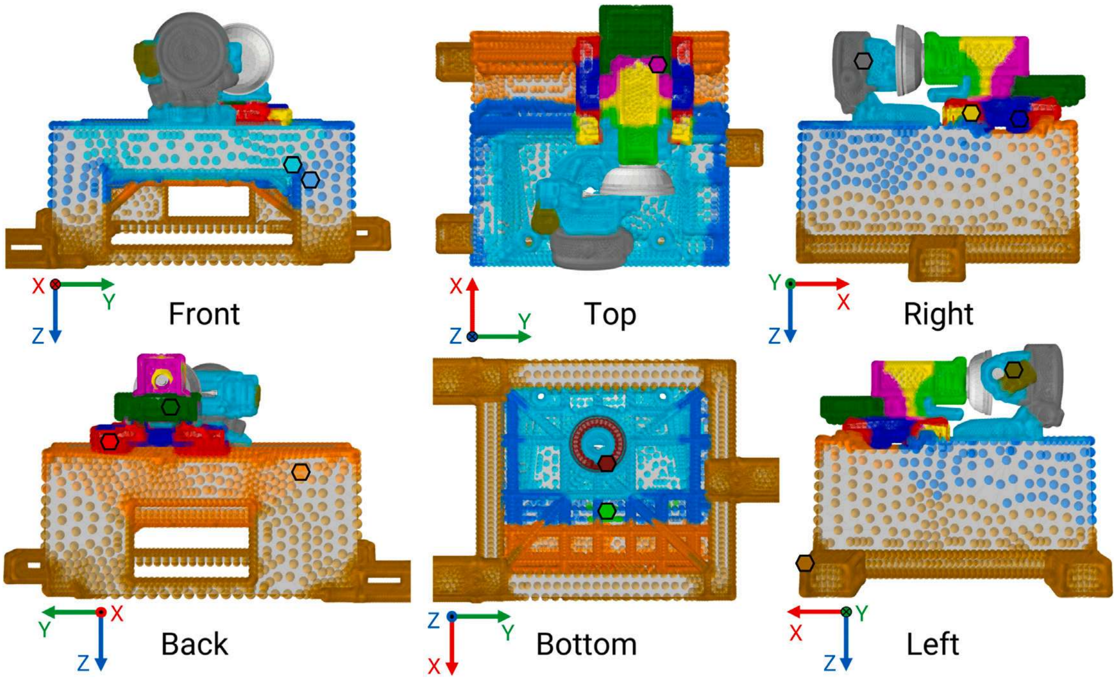

Fig. 6. K-means clustering on MT. A total of 13 clusters are evaluated according to the Calinski–Harabasz [50] criterion and the regions are marked with colored spheres on the corresponding nodes. The centroids that are used to select a sensor position are represented by a hexagon with a black contour. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

sensor assignments no longer change significantly, and the sensors within each cluster consistently exhibit similar temperature patterns over time.

K-means clustering can be hindered by its susceptibility to getting stuck in a local minima, which can lead to suboptimal cluster distribution. To overcome this issue, K-means++ [53] offers an improved option. It selects initial cluster centers based on probabilities proportional to each point's contribution to the overall potential. This approach could improve the clustering process by effectively distributing the initial centers as well as selecting the minimum from 25 new initializations created with a greedy K-means++ implementation. For reproducibility purposes, a random number seed of 42 is used.

All temperature measurement points of the DT are grouped into clusters. The determination of the number of clusters is carried out using the Calinski–Harabasz method [50], which assesses a specific number of sensors from all temperature measurements that are suitable for a given sensor range. The number of clusters also determines the final number of sensors selected. The sensor set will consist of the nodes nearest to the centroid for each cluster. The process of determining the positions according to the presented approach is summarized in Fig. 5. The simulation includes both the workpiece and the grinding wheel. Due to the high variability during the use in reality and the difficulty of temperature measurement, these components are excluded from the selection sets. It is important to note that the K-means clustering with euclidean distance assumes that clusters have roughly spherical shapes, the centroid however does not have to reside in the respective center depending on the presence of nearby clusters. A total of 13 sensors are chosen, and the resulting clusters with the respective centroids are indicated in Fig. 6. Each distinctly colored cluster represents an area from which one sensor, the centroid is selected. The MT bed has large cluster areas that divide the bed into distinct regions such as the ground, the grinding chamber, and the MT backroom. The workpiece holder is divided into four main clusters. The A-axis, C-axis and bed share a common cluster (light blue area), primarily because these areas are significantly influenced by the metalworking fluid used in the MT. However, the spindle stock contains the highest number of clusters. This is likely due to the presence of multiple heat sources in this region, including the spindle, X-axis and Y-axis drives, and heat generated by friction in the bearings. These heat sources create a complex temperature field, which is reflected in the clustering results. In Fig. 6, the cluster centers are marked by hexagons contoured in black. Typically, the centroids of spherical clusters are located in the center of the cluster. However, the clusters tend to be positioned at the edges of these regions in the cluster representation in Fig. 6. The X- and Y-axis consist of various clusters that are affected by the influence of different environments of the enclosure and heat sources from motors or due to friction that occurs between rotating parts and the linear rails.

3.3. Singular value decomposition

SVD is a technique to decompose a matrix $\mathbf{M} \in \mathbb{R}^{m\times n}$ into three distinct matrices: $\mathbf{U} \in \mathbb{R}^{m\times m}$, $\boldsymbol{\Sigma} \in \mathbb{R}^{m\times n}$, and $\mathbf{V}^T \in \mathbb{R}^{n\times n}$. The matrix $\mathbf{U}$ contains the left singular vectors, $\boldsymbol{\Sigma}$ is a diagonal matrix with singular values indicating the importance of each mode, and $\mathbf{V}^T$ contains the right singular vectors.

Geometrically, SVD can be interpreted as the transformation of a unit sphere. Under a linear transformation, represented by the matrix $\mathbf{M}$, the image of a unit sphere becomes a hyperellipse. This is illustrated in the right part of Fig. 7, which demonstrates the transformation of a unit circle into an ellipse. The matrix $\mathbf{M}$ is decomposed into two rotations, represented by the matrices $\mathbf{U}$ and $\mathbf{V}^T$, and a scaling along the coordinate axes by the singular values σ_1 and σ_2 of the matrix $\boldsymbol{\Sigma}$. The singular values in the matrix $\boldsymbol{\Sigma}$ are generally sorted in descending order. By selecting only the top n singular values and their corresponding vectors, the data can be approximated with fewer dimensions. This reduced-dimensional representation captures the relevant patterns in the data while filtering out less important details.

In this approach, SVD is employed to identify a set of temperature sensors, as illustrated in Fig. 7. The evaluation of sensor positions is based solely on temperature data from the simulation, as in the previously presented centroid k-means approach. All measurements are considered, and SVD is utilized to compress the temperature field, retaining only a limited number of strategically chosen measurement points. In the SVD process, the number of modes and the number of

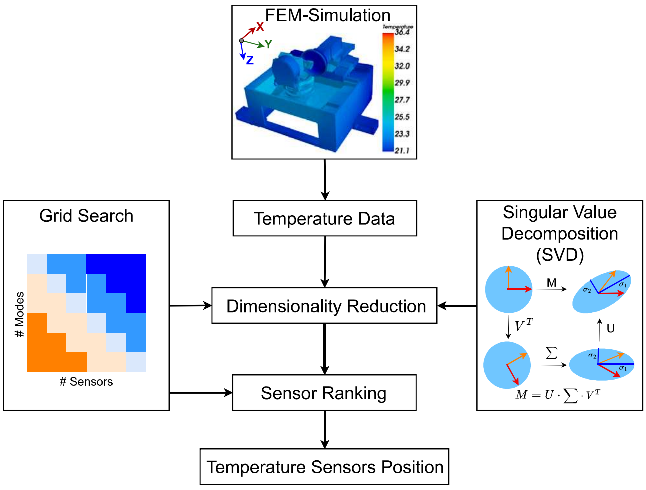

Fig. 7. Flowchart of the SVD approach. Temperature data is obtained from the FEM simulation, which is then reduced by SVD to obtain a set of distributed sensor positions on the MT.

sensors for selection can be adjusted. The modes refer to the basis functions or principal components derived from the SVD of the temperature data. These modes capture the most significant patterns in the data. For example, if the number of modes is set to 20, the model will consider the top 20 SVD modes. This implies that the data is approximated by combining the specified number of modes, since these modes capture the most important variations. However, it is crucial to note that these parameters are interdependent. The choice of the number of modes can influence the selection of measurement points, leading to variations in the evaluated positions. To address this, grid search is conducted to optimize both of these parameters, exploring combinations of the number of basis modes and the number of sensors. The Mean Squared Error (MSE) of the reconstructed temperature field is computed for each combination. The results of the grid search indicated that a practical rule of thumb is to select an equal or higher number of sensors than the number of modes. This systematic approach ranks all nodes and assigns temperature sensor positions based on the calculated singular values.

Fig. 8 shows the six temperature sensors selected by applying SVD with cross-validation to the first simulation data set. These positions can then subsequently be used to train and validate a thermal compensation model.

3.4. Group LASSO for identifying suitable temperature sensor positions

LASSO is a widely used technique in regression analysis that integrates variable selection and regularization. Tibshirani et al. [54] introduced the LASSO to augment traditional regression by incorporating L1-regularization, as described by Eq. (4).

$$\hat{\beta}_\lambda = \arg\min_{\beta} \|\mathbf{y} - \mathbf{X}\beta\|_2^2 + \lambda \sum_{j=1}^{p} \left|\beta_j\right| \tag{4}$$

$\mathbf{y}$ represents the thermal error to be modeled. The LASSO method is applied to predict this thermal error based on data collected from various temperature sensors. The matrix $\mathbf{X}$ contains all the temperature measurements from sensors that are used to estimate $\mathbf{y}$. The variable p denotes the total number of parameters or features in the model, which is equal to the number of potential temperature measurement points considered. The inclusion of L1-regularization removes irrelevant elements from the model parameter vector, denoted as $\boldsymbol{\beta}$. By pruning these parameters, the corresponding model inputs are also eliminated, thereby identifying the selected inputs and their associated parameters in one operation. The parameter $\lambda \geq 0$ governs the significance of the regularization term in the optimization and dictates the number of selected features and parameters in the resulting regression model.

$$\hat{\beta}_\lambda = \arg\min_{\beta} \|\mathbf{y} - \mathbf{X}\beta\|_2^2 + \lambda \sum_{g=1}^{C} \left\|\beta_{I_g}\right\|_2 \tag{5}$$

The Group LASSO extends the LASSO approach by eliminating groups of features, as outlined in Eq. (5). In this context, a group comprises a temperature sensor and its lagged instances, encompassing its historical states. This strategy allows the identification of groups that have a strong relationship with the displacement data, resulting in a reduced set of temperature sensors. The variable C is equal to the total number of groups or temperature sensors. Each group contains one sensor and its lagged states, preventing, for example, the use of many sensors but only certain lagged states of each sensor. Considering the time dependency of the inputs increases the amount of data that needs to be processed. This results in longer computation times for the Group LASSO to identify suitable positions. Initially, around 185,000 positions are considered. Adding three different lagged time states further multiplies the total data samples analyzed by Group LASSO by a factor of four.

3.5. Group LASSO with SVD preselection

To streamline the evaluation of temperature sensor locations amidst numerous potential options and combinations, a two-step selection process has been developed, aiming to reduce computational complexity more applicable to extensive datasets, as illustrated in Fig. 9. In the first step, the pool of potential sensor locations is reduced using SVD, since it is very efficient in handling a large amount of data. This helps to reduce the large number of temperature sensors to a smaller subset. This selection should help to thin out the huge amount of possible sensor positions as a coarse pre-selection. Notably, this reduction retains a significant number of sensors for consideration. Subsequently, the Group LASSO is set up to introduce a finer and more accurate selection of sensors using the final model structure, since the computational effort is reduced by the pre-selection step. For this purpose, lagged time is incorporated to include past temperature data and group them for each sensor, thereby improving the model's ability to capture temporal patterns. The Group LASSO is applied to the time-lagged data set, resulting in a selected set of sensor locations. The introduction of the pre-selection step helps the Group LASSO to handle the large amount of possible sensor positions and to focus only on a few coarse pre-selected positions. However, too many pre-selected positions are available for direct modeling, since we reduce the number of potential positions from 185,000 to only around 800. Despite this reduction, a vast amount of sensors remains when considering the constraints of manpower and time that is needed if the selection is carried out on the physical MT. Following the two-step selection procedure presented, the sensor setup is shown in Fig. 9(c). This combined approach allows the best of both approaches, the computational efficiency of the SVD chooses a minimal representation of the thermal effects. The group LASSO, albeit computationally intensive, allows for the sensors to be chosen that are the actually the best at predicting the thermomechanical behavior. A pure LASSO approach is much more computationally intense than with the pre-selection and can perform worse at it is fitted closer to the training data and less to fit a general thermal representation of the system. Therefore this novel approach allows the efficient selection of ideal temperature measurement locations based on the digital twin.

4. Compensation model for machine tools

To effectively compare various sets of evaluated sensors, the Autoregressive with exogenous input (ARX) model proves to be a valuable

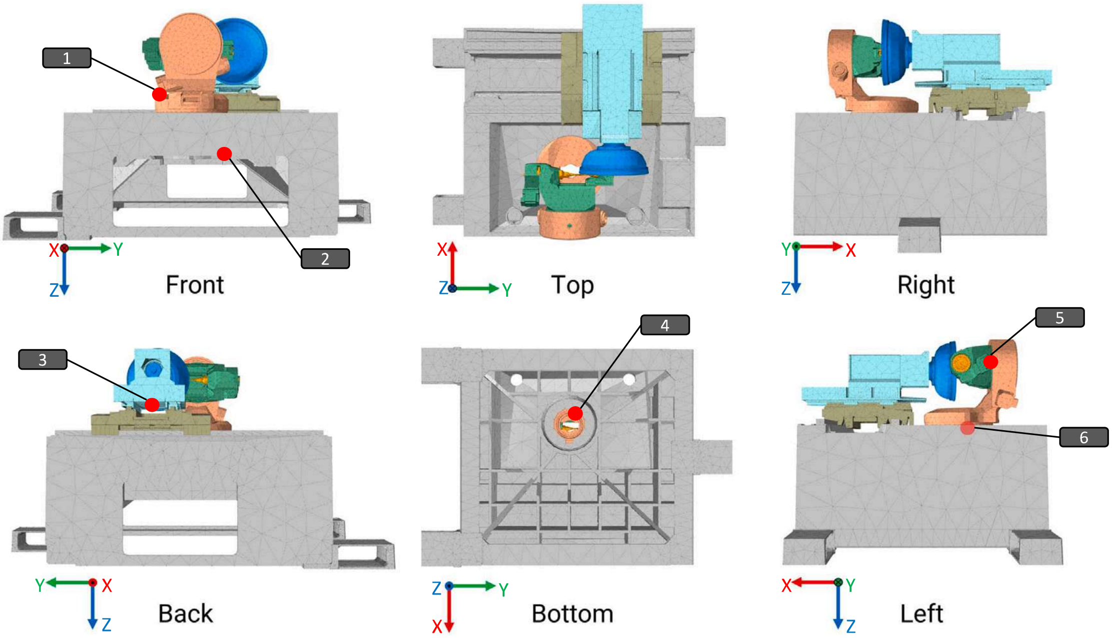

Fig. 8. Six temperature sensor positions proposed by applying dimension reduction of the temperature field using SVD.

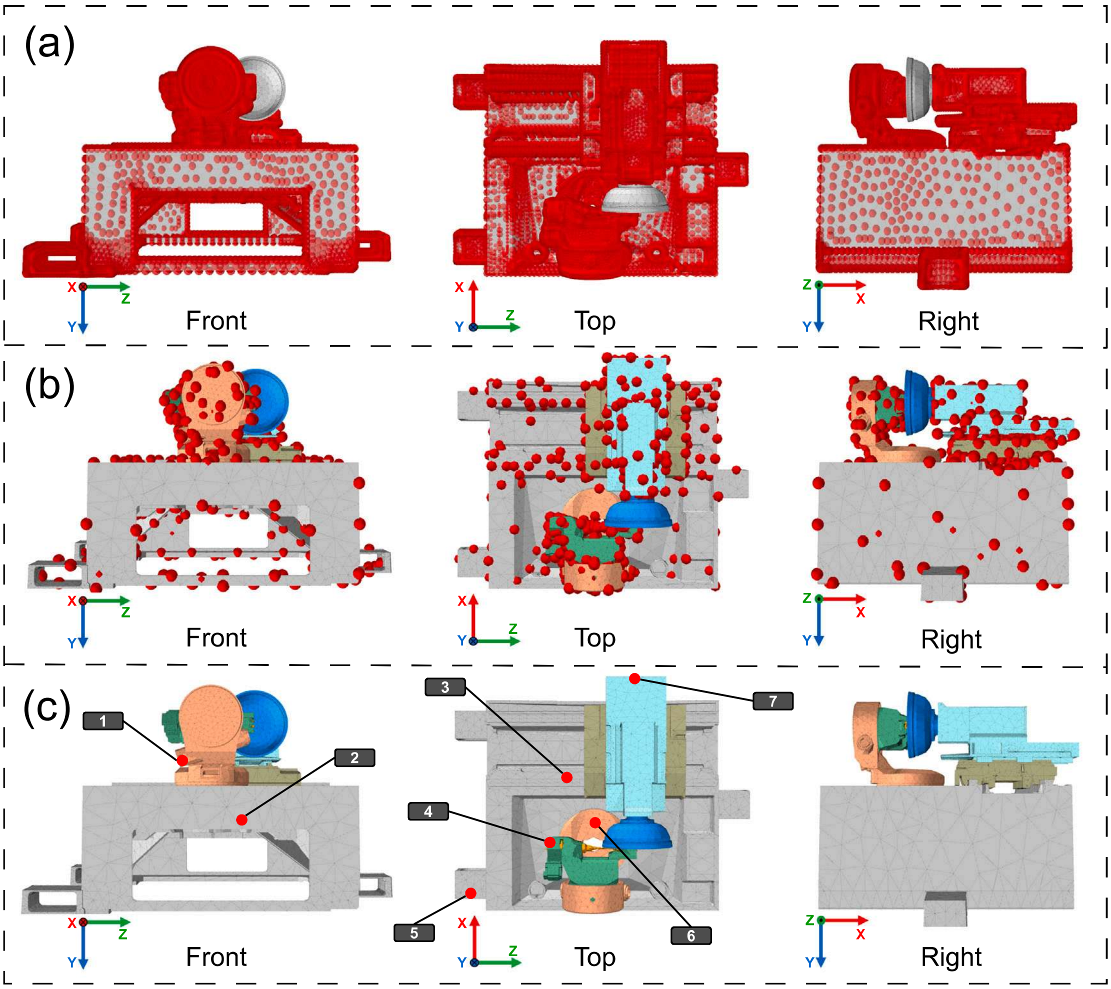

Fig. 9. The top image (a) shows all possible sensor locations, so all surface mesh elements with the exclusion of the grinding wheel as well as the workpiece. Due to the large number of possibilities the numbers are reduced by SVD as a pre-selection which is shown in (b) and at last the Group LASSO is applied yielding the final sensor set containing 7 temperature sensors shown in (c).

tool. This model is adept at predicting thermal errors and subsequently compensating for them based on forecasted values.

$$y[k]+\sum_{i=1}^{n_a}\beta_i\cdot y[k-i]=\sum_{m=1}^{M}\sum_{n=0}^{n_b(m)}\beta_{m,n}\cdot u_j[k-n] \tag{6}$$

The ARX model, a linear time-invariant dynamic system, is represented by a weighted sum and follows a multiple-input single-output (MISO) configuration. So the output $y[k]$ at a timestep k is a thermal error of one type/one direction. The autoregressive component is considered by n_a steps of past predictions using the model parameters fitted to this β_i. The inputs, in this case measurements of temperature sensor m at a timestep k up to $n_b(m)$ previous steps, is multiplied with the respective parameter $\beta_{m,n}$. In this setup, each thermal error of the MT is predicted by a distinct independent model. The fundamental idea is to employ the ARX model for thermal error prediction through simulations, utilizing temperature sensors evaluated from different methodologies. The simulation employs identical load cases to evaluate and compare sensor positions across all datasets obtained from diverse sensor setups. The essence of the approach lies in comparing the compensation performance of each sensor set to discern the optimal sensor positions. Presently, assessing sensor positions through engineering knowledge remains the standard for installing temperature sensors even in research, making it the benchmark against which other sensor sets are evaluated.

The proposed approach of the digital twin to identify optimal temperature sensor positions can be used in combination with any model both, physics and data driven for subsequent compensation. Instead of the ARX model also neural network based approaches suitable for time-series data such as LSTM models can be successfully used for thermal compensation. The increased suitability of the sensor positioning will benefit all model types as the input data allows a better reconstructing of the underlying thermal effect as well as the required observation of the resulting thermomechanical deformation.

4.1. Experimental setup of the investigated machine tool

To demonstrate the applicability of the developed sensor set and thermal error compensation using real-world production machines, not only the digital twin but also the physical MT is investigated. The 5-axis MT has the following kinematic chain according to ISO 10791-1:2015 [55]: $V[w-B'-A'-C'-b-Y-X-(A)-t]$. In order to measure the thermal error on the MT a 3D touch trigger probe is installed next to the grinding spindle. A repeated measurement of the workpiece holder allows for the deviations in X-, Y- and Z-direction to be measured in the same way as in the simulations associated with the digital twin. Furthermore, a total of 24 PT100 RTD temperature sensors are installed on the MT. The sensors are installed as shown in Fig. 4. Two additional sensors are positioned along the C-axis, as recommended by the SVD & Group LASSO approach depicted in Fig. 9. This allows for comparison between the best-optimized set based on SVD & Group LASSO and the engineering knowledge-based sensor placements. The temperature measurements are executed every 5 s while the displacement is only measured every 188 s with intermittent air grinding of a workpiece to introduce thermal load.

5. Results and discussion

This section compares the performance of different temperature sensor sets used for thermal error compensation. First, the validation set from the simulations is analyzed. Then measurements carried through on the actual MT are used to compare the manually selected set of 22 sensors with the optimized set of 7 sensors, chosen using LASSO and SVD.

With two exceptions, the FE-selected set is regarded as a subset of the manually placed sensors, as exact placement on the real machine is constrained by accessibility and cannot match the mesh element resolution.

Table 3
Compensation model performance on the validation data set from simulation in the average RMSE reduction in X-, Y- and Z-direction after the model is trained for 160 h. Different sensor sets are compared that are selected based on the previously introduced methods.

Training time 160 h	#Sensors	#Model param.	E_X [%]	E_Y [%]	E_Z [%]
Eng. Knowledge	17	35	73	97.8	83.5
Centroid (K-means)	13	25	64.5	97.6	83.1
SVD 7 Modes	10	19	79.2	97.6	91.5
SVD 5 Modes	6	19	74	94.5	69.3
Group LASSO & SVD	7	22	81.1	93.6	77.2

5.1. Comparison of compensation models based on different positioning approaches

Table 3 summarizes the performance of compensation models evaluated on the validation data set presented in Fig. 3 which was generated by a simulation. The compensation model for the engineering knowledge set applied to this data is shown in Fig. 10. The root mean square error (RMSE) serves as the primary metric for assessing the accuracy of thermal error compensation in the X-, Y-, and Z-directions. The first half of the validation data set is used to train the ARX model parameters, while the second half is reserved exclusively for evaluating model performance. RMSE reduction (E_X, E_Y, E_Z) is expressed as the percentage improvement in thermal error relative to the uncompensated state, with higher values indicating better compensation. A reduction of 100% corresponds to zero deviation from the initial state over the entire evaluation period. The compensation model based on engineering knowledge utilizes with 17 the most temperature sensors, making it the largest model. Although its performance in the X-direction (73%) is lower than that of most other approaches, it achieves the highest reduction in the Y-direction (97.8%) and average results in the Z-direction (83.5%). This indicates that a suitable sensor set can also be determined through manual selection, albeit at the cost of additional effort and resources. The centroid-based approach using k-means clustering uses 13 sensors, reducing RMSE by 64.5% in the X-direction which is the worst in the most critical dimension despite average values in Y- and Z-direction. The SVD-based approaches, configured with 7 and 5 modes, strike a balance between model complexity and performance. The 7-mode SVD configuration requires 10 sensors but consistently outperforms the 5-mode SVD setup which uses just 6 sensors, particularly in the Z-direction (69.3%). This highlights the trade-off between reducing sensor count and maintaining model accuracy (see Fig. 10). The Group LASSO & SVD approach, employing only 7 sensors and 22 parameters, demonstrates the highest accuracy in the X-direction (81.1%), while maintaining competitive RMSE reductions in the Y-direction and in the Z-direction. Given that X-direction errors are the largest and most critical for grinding MTs, as they directly impact the radius and thereby double their effect on the workpiece diameter, this approach emerges as the most effective overall. These findings emphasize the pivotal role of optimized sensor placement strategies in thermal error compensation and show the benefit in the use of the hybrid Group LASSO & SVD approach. Advanced techniques, such as Group Lasso and SVD, deliver substantial improvements over traditional methods like manual engineering knowledge-based placement. By achieving significant error reduction with fewer sensors and reduced model complexity, these methods offer scalable, cost-effective, and robust solutions for precision manufacturing.

It has to be considered however that the sensor positioning has been done only on the first simulative data set which therefore is limited by the applied boundary conditions. Varying these infinitely is not computationally feasible as both the simulation of the entire machine as well as the sensor positioning using Group LASSO are demanding operations. The use of a separate load scenario for validation confirms however that the selected set is applicable to a broad range of typically occurring machine uses and environments.

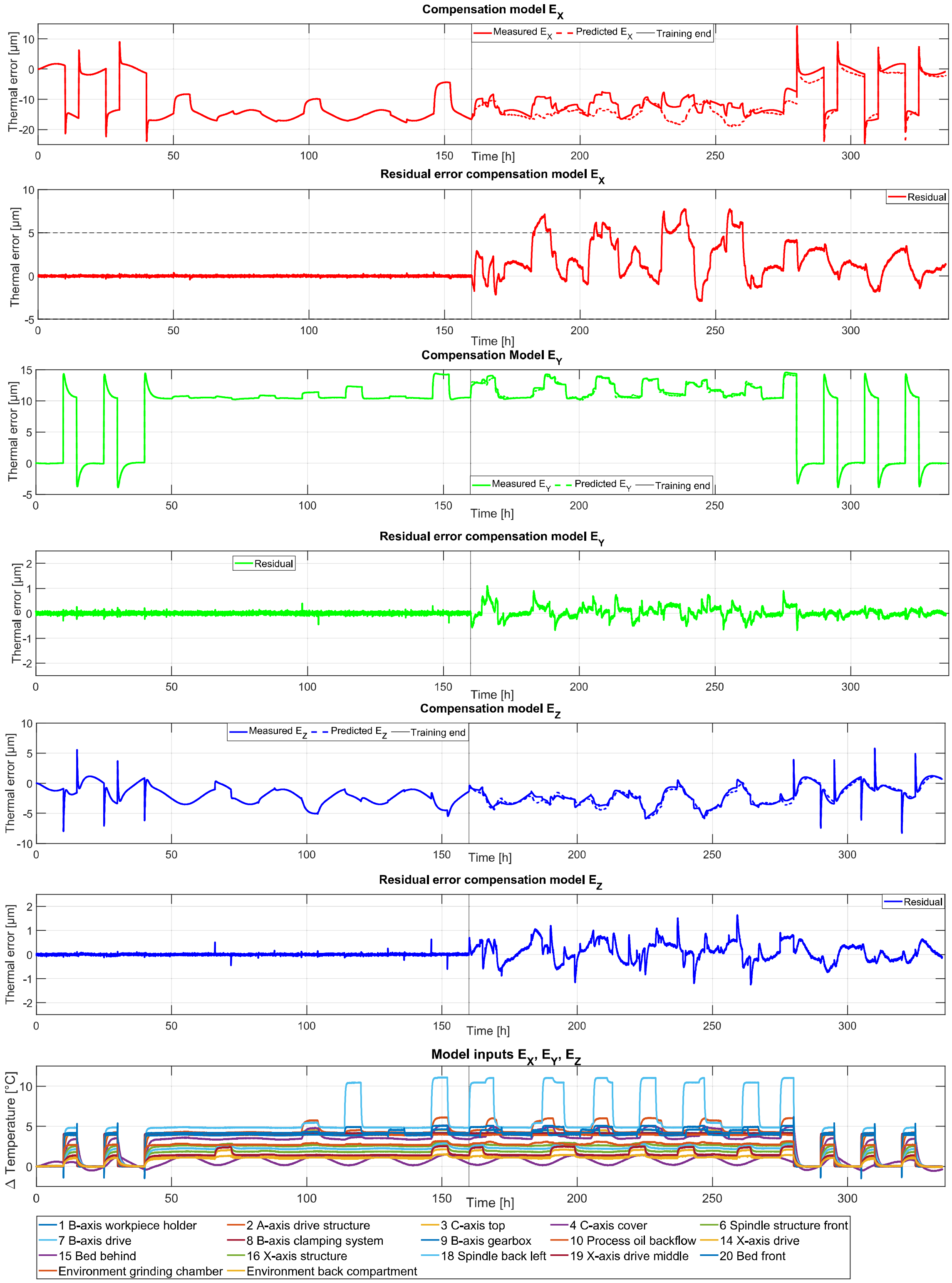

Fig. 10. Compensation models for the validation simulation data set of the X-, Y- and Z-Error. All temperature sensors of the manually placed set, according to engineering knowledge, are used as model inputs for all three compensation models.

5.2. Validation measurements on a real grinding machine tool

To validate the selected sensor set and assess the compensation approach on the actual MT, a series of experiments totaling 108 h is conducted. Thermal errors in both the X- and Y-direction are recorded at intervals of 188 s, along with temperature readings from all sensors in both the manually chosen and optimized sensor sets as described in Section 4.1. This validation procedure comprises seven discrete experimental sessions, each separated by intervals to examine the MT's thermal behavior under varying conditions. These interruptions, indicated by red-shaded areas in the figure, are not to scale; in some cases, interruptions span multiple days.

Table 4 compares the manually positioned sensors, with and without input selection, to those of the Group LASSO & SVD set. The

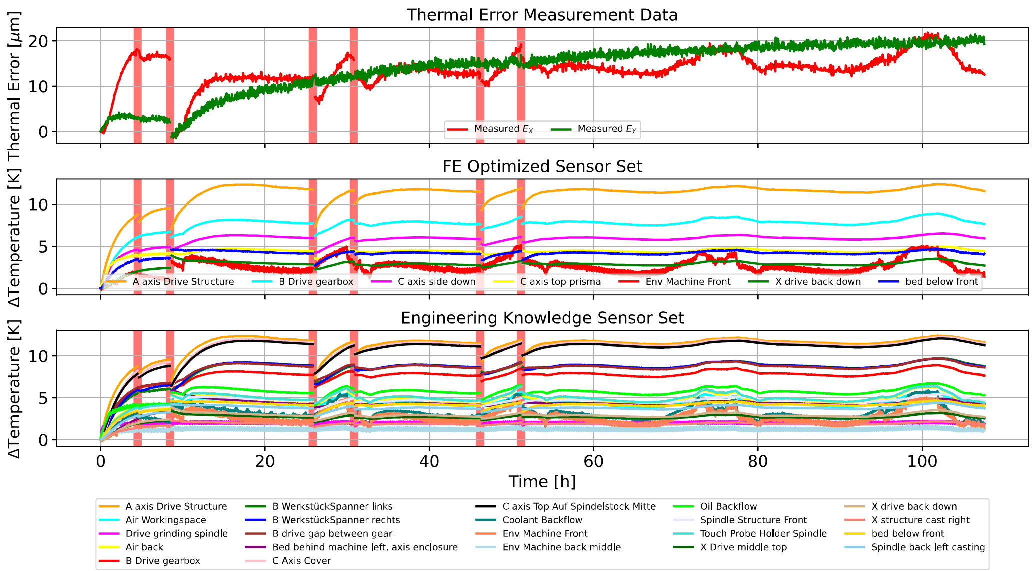

Fig. 11. Thermal error measurement data. The measurements along the X- and Y-direction of the probe are averaged. The second subplot shows the temperatures measured by the sensor set chosen based on SVD & Group LASSO. The third subplot shows all temperature sensors based on the manual positioning based on engineering knowledge. The red areas represent not to scale machine interruptions which change the thermal state. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 4
Compensation model performance on real measurement data in the average RMSE reduction in X- and Y-direction after the model is trained for 52 h.

Training time 52 h	#Sensors	E_X comp [%]	E_Y comp [%]
Engineering knowledge	22	91.9	73.4
Engineering knowledge with input selection	5	81.1	67.8
Group LASSO & SVD	7	85.2	75.5

exact same compensation model architecture and hyperparameters are used and only the input set is varied. It can be observed that the average RMSE reduction is quite high for both sets indicating a suitable compensation model can be found for both input sets. In X-direction the error can be compensated slightly worse with the Group LASSO & SVD set as optimized by using the simulation, as only 7 instead of 22 temperature sensors are available. In Y direction the accuracy is generally slightly reduced compared to the X-error due to the lower magnitude and slightly larger influence of the measurement noise on the overall measurement. In this case, the Group LASSO & SVD set slightly outperforms the manually positioned set. If state of the art input selection is applied to further reduce the 22 sensors of the engineering knowledge 5 sensors are chosen. This however performs the worst of the sensor sets but still in an acceptable range. The full set without input selection also performs well because the sensors are positioned with great care to prevent or reduce collinearity and the chosen model architecture is robust. This indicates that both sets are suitable and generally chosen well, but the optimized set is much more efficient and well-conditioned compared to the manual approach, even when carried through by a team of experts, as in this case. Simply applying input selection on manually chosen physical sensors on the MT is also slightly worse than using the simulation to find the optimal set. This indicates that using the digital twin framework helps finding the overall most suitable temperature sensors for thermal error compensation not only in simulation but also in real-world conditions.

Fig. 11 shows the measured thermal error in the X- and Y-direction. It can be observed that a total thermal error of up to ∼20 μm is reached on the MT. The X- and Y-error show very different characteristics. However, both clearly correlate with some of the measured temperatures. Interrupting the MT resets the temperature depending on the duration of the interruption and, therefore, also typically reduces the thermal error. Especially in X-direction it reacts faster after such restarts. The second and third subplot show the temperatures of the SVD & Group LASSO and engineering knowledge data set, respectively. A total temperature variation of more than 12 °C can be observed. Some of the sensors are contained in both sets as accessibility restrains lead to often only one feasible sensor location.

Fig. 12 shows the ARX compensation model introduced in Section 4 applied to the real measurement data of the physical MT. The first 50%, so 54 h of measurement data, are used for the model training, while the subsequent measurement is only used to verify the model effectiveness and, therefore, the validity of the sensor set. After the model is trained it is applied to predict both the training and validation set. The first subplot shows both the measured and predicted error in X-direction. The second subplot shows the residual error of the compensation model so the difference between the actual measurement value and the model prediction. It can be observed that during the training phase, the prediction and measurement match very well. The residual is always below 2 μm and centered around zero. The behavior appears erratic, similar to the total measurement uncertainty. This is most likely due to external vibrations, the measurement uncertainty of the touch probe itself and the slight time variation between temperature measurements, and the fact that the measurement cycle is not instantaneous. After the training phase, however, this behavior slightly changes. The noise is still present, but now there is a slight constant offset, ranging between 0–5 μm between the prediction centerline and the measurement, indicating a systematic deviation. This could, for example, be caused by slippage in the system leading to a loss of repeatability or thermal effects not observed in the training phase. However, the offset is small in relation to the overall thermal error, which is reduced from an RMSE of 7.1 to 1.1 μm, an 85.2% reduction. This is achieved using only the 7 temperature sensors, shown in the third subplot, as exogenous model inputs. Such a reduction is crucial for manufacturing high-precision components, where tolerances often measure only a few μm or less. In some cases, the machine kinematics can amplify certain errors. For example, the width of a feature produced at an angle may be disproportionately affected due to the magnification of positional deviations along the normal distance to the machining surface.

Fig. 13 similarly shows the error in Y direction and the respective model performance. The measurement and then model noise are similar to that of the X-direction, never reaching 2 μm. However, after the

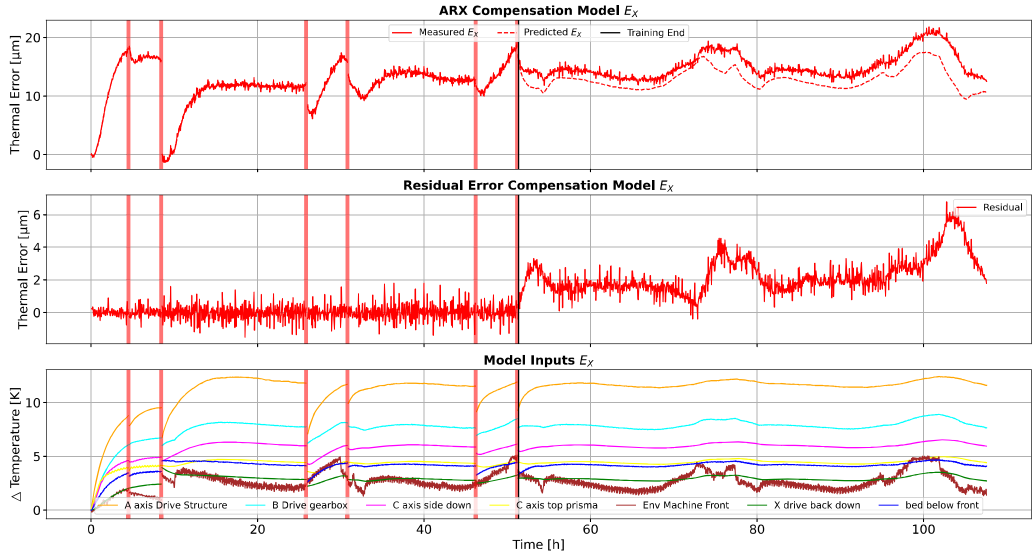

Fig. 12. Compensation of E_X on real measurement data on the MT. A measurement of 108 h is performed. The red areas represent not to scale machine interruptions which change the thermal state. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

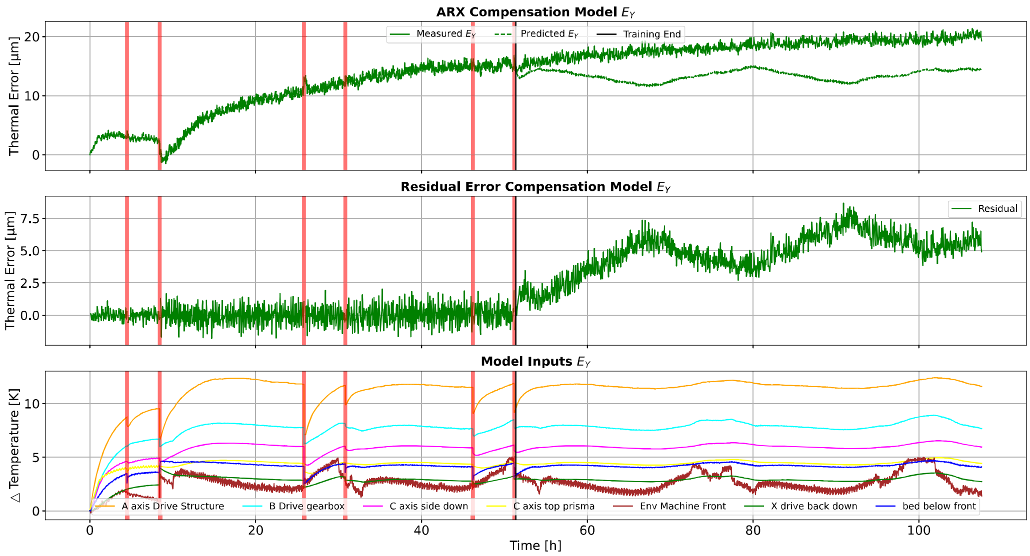

Fig. 13. Compensation of E_Y on real measurement data on the MT. A measurement of 108 h is performed. The red areas represent not to scale machine interruptions which change the thermal state. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

model training, the deviation between the prediction and measured value appears more constant, with two "day cycles" being observed. This indicates that the temperature set leads to slightly worse performance as one temperature sensor for the environment might be too little. The measured environmental temperature sensor fluctuated by more than 5 °C. The machine might not be homogeneously affected by one sensor but rather different environmental effects which is why a residual environmental component remained. However the overall reduction of 75% is still considered very high and the model appears to be robust.

This compensation enables the elimination of warm-up phases for most components, with tolerances of a few μm, which were previously required to ensure no scrap was produced or higher measurement or re-grinding efforts. Concurrently, a higher robustness to environmental temperature fluctuations is achieved. This enhancement in energy efficiency is achieved by twofold means: first, the machine's utilization and accuracy is enhanced, leading to increased productivity and reduced energy expenditure and scrap production. Concurrently, the energy consumption for the climate-controlled production can be reduced. Consequently, thermal compensation with an optimized sensor set demonstrates avenues to augment the sustainability of the manufacturing process through the implementation of digital twins.

6. Conclusion and outlook

This paper introduced a digital twin-based approach for optimizing thermal error compensation in MTs through autonomous and intelligent sensor positioning. By using the digital twin framework to define optimal sensor locations, the proposed method achieved robust thermal error compensation with a significantly reduced number of sensors. Our results show that with only seven sensors, this approach can match or exceed the accuracy of manual placement using up to 22 sensors. It has been demonstrated that the proposed method clearly outperforms traditional input selection approaches that rely on experimental sensors. The proposed method enables the integration of arbitrary model parameter variations and boundary conditions during the selection process, thereby eliminating the need for extensive experimental data.

This flexibility enhances the accuracy and robustness of sensor placement and significantly reduces costs and time associated with physical testing. This reduction not only enhances model robustness and sustainability but also minimizes costs associated with sensor installation and maintenance and reduces the risk of sensor failure leading to model impairment. Furthermore, the task of input selection becomes easier to perform on the edge as the selection set is significantly improved and smaller.

Validation on a real 5-axis precision grinding MT further demonstrated that the optimized sensor set achieves high compensation accuracy across multiple operational conditions, significantly reducing thermal errors in the X- and Y-directions by 85% and 75% respectively. This performance underscores the effectiveness of selecting optimal temperature sensor positions using SVD and Group LASSO. Integrating these methods within a digital twin framework enables broad applicability across diverse scenarios that are challenging to replicate in a development environment, accelerating the design process and enhancing model precision well before a physical prototype is constructed.

Future work will focus on expanding the transferability of this digital twin-based approach to different machine configurations and operational contexts, investigating how machine-specific models might be adapted across diverse systems. This will help prove the applicability for machines in any environment. Incorporating active learning and self-learning algorithms could enhance model adaptability, enabling real-time adjustments to the used sensor set and thermal compensation based on evolving machine conditions. These sensors also allow other uses such as insights for predictive maintenance, process optimization and demand oriented cooling control. Ultimately, this research paves the way for more intelligent, autonomous, and sustainable thermal compensation in high-precision manufacturing, positioning the use of digital twins and compensation as a cornerstone of future MT design and control strategies.

CRediT authorship contribution statement

Sebastian Lang: Writing – original draft, Software, Investigation, Data curation, Conceptualization. **Mario Zorzini:** Writing – review & editing, Software, Data curation. **Stephan Scholze:** Writing – review & editing, Project administration, Funding acquisition. **Josef Mayr:** Writing – review & editing. **Markus Bambach:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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